

|                                                                                      |                                                                  |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Program: Diploma in Electrical Engineering&amp; Electrical Vehicle Technology</b> |                                                                  |
| Course Code: <b>6412A</b>                                                            | Course Title: <b>Fundamentals of Electric Vehicle Technology</b> |
| Semester: <b>6</b>                                                                   | Credits: <b>4</b>                                                |
| Course Category: <b>Open Elective Course</b>                                         |                                                                  |
| Periods per week: <b>4 (L:3 T:1 P:0)</b>                                             | Periods per semester: <b>60</b>                                  |

### Course Objectives:

- To understand basic concepts and types of EV's
- To understand about the charging systems of batteries
- To understand the basics of Power Electronics and working of EV motors
- To understand the EV communication protocols, technology and charging methods

### Course Prerequisites:

| Topic                                             | Course code | Course Title                                          | Semester |
|---------------------------------------------------|-------------|-------------------------------------------------------|----------|
| Basic knowledge about storage batteries           | 2411        | Introduction to Electric Vehicles & Elementary Theory | 2        |
| Basic Knowledge of batteries and charging systems | 3411        | Electric Vehicle Batteries & Charging Systems         | 3        |

### Course outcomes:

On Completion of this course, the student will be able to:

| CO n | Description                                                         | Duration (Hours) | Cognitive level |
|------|---------------------------------------------------------------------|------------------|-----------------|
| CO1  | Outline the basic concepts and types of EV's                        | 15               | Understanding   |
| CO2  | Understand about the charging systems of batteries                  | 15               | Understanding   |
| CO3  | Understand the basics of Power Electronics and working of EV motors | 14               | Applying        |
| CO4  | EV communication protocols, technology and charging methods         | 14               | Understanding   |
|      | Series Test                                                         | 2                |                 |

## CO-PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             | 3   |     |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

## Course Outline:

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                     | Description                                                                    | Duration (Hours) | Cognitive Level |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Outline the basic concepts and types of EV's</b>                            |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the need for hybrid and electric vehicles                              | 4                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the components and working principles of electric and hybrid vehicles. | 4                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                                               | Identify the types of hybrid Electric vehicle                                  | 3                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the life cycle assessment of EV's                                      | 4                | Understanding   |
| <b>Contents:</b><br><br>Need for hybrid and electric vehicles, Historical background, Future trends in EV - Government Schemes and Progress: FAME-1, FAME-2-State Policies, subsidies, and incentives.<br>Main components and working principles of electric and hybrid vehicles.<br>Types of HEV - series, parallel, series-parallel and Complex. Comparative study of EV & Conventional vehicles - Life cycle assessment of EV's. |                                                                                |                  |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Understand about the charging systems of batteries</b>                      |                  |                 |
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the configurations of EV                                               | 4                | Understanding   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                        |   |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---|---------------|
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Identify the types of charging systems                                                 | 3 | Understanding |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Explain the types of batteries                                                         | 4 | Understanding |
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Illustrate various parameters of battery                                               | 4 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Series Test I                                                                          | 1 |               |
| <b>Contents:</b><br><b>Configurations of EV-</b> – Pure Electric vehicle (PE)-Battery Electric vehicles (BEV)-Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) – Plug-in Hybrid Electric vehicle (PHEV)<br><b>Charging System</b> - On board & OFF board charging, domestic charging, fast charging, public charging.<br><b>Types of batteries</b> - Lead Acid Battery, Li-ion battery-Battery Management System, Basic principle of ultra-capacitor<br><b>Battery parameters:</b> Battery parameters: Physical Dimensions, Voltage and current rating, Capacity, and power, C Rate, Battery Efficiency, Energy Density, Power Density, State of charge (SOC), Depth of discharge (DOD), State of Health (SOH) |                                                                                        |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Understand the basics of Power Electronics and working of EV motors</b>             |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Outline the EV motor types and requirements                                            | 4 | Understanding |
| M3.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Explain the basics of power electronic components related to EV's                      | 4 | Understanding |
| M3.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Explain power split devices for HEV                                                    | 3 | Understanding |
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Illustrate the types of motor controllers                                              | 3 | Understanding |
| <b>Contents:</b><br>Requirements of EV Motors - Basic idea of types of EV motors: AC motor, switched reluctance motor, Brushed & brushless DC motor<br>Power electronics related to EV – Rectifiers(DC-AC converters), Inverters (DC-AC converters), DC/DC converters(choppers)-sensors (Temperature , MEMS, Battery current) - Power Split devices for Hybrid Vehicles.<br>Motor Controllers- Variable resistor type , Pulse Width Modulation.                                                                                                                                                                                                                                                                                 |                                                                                        |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>EV communication protocols, technology, charging methods &amp; grid integration</b> |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Explain the charging methods of EV & HEV                                               | 2 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Explain about wireless power transfer technology of EV charging                        | 4 | Understanding |

|                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                         |   |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---|----------------|
| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                           | Explain the various protocols used for EV communication | 4 | Understanding  |
| M4.04                                                                                                                                                                                                                                                                                                                                                                                                                           | Outline the concept of E- mobility and electrification  | 4 | ,Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                 | Series Test II                                          | 1 |                |
| <b>Contents:</b><br><b>Charging of EV and HEV:</b> Battery swapping, Smart charging, Wireless charging- EV grid integration<br><b>Wireless power transfer (WPT)</b> technique for EV charging.<br><b>Protocols and communication:</b> Specification of open charge point protocol (OCCP 1.6/2.0) - Bharat DC001 & AC001 Charger specification-CAN communication<br><b>E-mobility and electrification</b> - scope and challenges |                                                         |   |                |

#### Text/Reference:

| T/R | Book Title/Author                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------|
| R1  | Jack Erjavec, Jeff Arias, “Hybrid, Electric and Fuel-cell Vehicles”, Cengage Learning India Pvt Ltd. New Delhi. |
| R2  | S. Onori, L. Serrao and G. Rizzoni, “Hybrid Electric Vehicles: Energy Management Strategies”, Springer, 2015    |
| R3  | T. Denton, “Electric and Hybrid Vehicles”, Routledge                                                            |
| R4  | A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi (Ed.                                  |
| R5  | C. Mi, M. A. Masrur and D. W. Gao, “Hybrid Electric Vehicles: Principles and                                    |

#### Online Resources:

| Sl. No | Website Link                                                                                                                                          |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | <a href="https://nptel.ac.in/courses/108/103/108103009/">https://nptel.ac.in/courses/108/103/108103009/</a>                                           |
| 2      | <a href="https://nptel.ac.in/courses/108/106/108106170">https://nptel.ac.in/courses/108/106/108106170</a>                                             |
| 3      | <a href="https://www.sciencedirect.com/science/article/pii/S2095756420300647">https://www.sciencedirect.com/science/article/pii/S2095756420300647</a> |
| 4      | <a href="https://www.metisengineering.com">https://www.metisengineering.com</a>                                                                       |
| 6      | <a href="https://onlinecourses.nptel.ac.in/noc20_ee99/preview">https://onlinecourses.nptel.ac.in/noc20_ee99/preview</a>                               |